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5.2.2 LU factorization: The right-looking algorithm

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other, the corresponding parts. So what we conclude is that α_{11} must be equal to ϵ_{11} , a_{21} , sorry, a_{12} transpose must be equal to u_{12} transpose, a_{21} must be equal to l_{21} times ϵ_{11} , which, by the way, we could have written as ϵ_{11} times l_{21} because a vector times the scalar is the same as a scalar times a vector. And finally, A_{22} is equal to -- let's see-- l_{21} times u_{12} transpose plus l_{22}

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Reading assignment

1/1 point (graded)

Read Unit 5.2.2 of the notes. [\[LINK\]](#)

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Homework 5.2.2.1

1/1 point (graded)

Choose the approximate cost incurred by

$A = \text{LU-right-looking}(A)$

$A \rightarrow \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right)$

A_{TL} is 0×0

$\text{while } n(A_{TL}) < n(A)$

$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c|cc} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right)$

$a_{21} := a_{21} / \alpha_{11}$

$A_{22} := A_{22} - a_{21} a_{12}^T$

$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \leftarrow \left(\begin{array}{c|cc} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right)$

endwhile

when applied to an $n \times n$ matrix.

- ☐ $2n^3$
- ☒ $\frac{2}{3}n^3$
- ☐ $\frac{1}{3}n^3$



Solution:

For a complete solution, see [the notes](#).

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Answers are displayed within the problem

Homework 5.2.2.2

1/1 point (graded)

Choose the approximate cost incurred by

$A = \text{LU-right-looking}(A)$

$A \rightarrow \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right)$

A_{TL} is 0×0

$\text{while } n(A_{TL}) < n(A)$

$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c|cc} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right)$

$a_{21} := a_{21} / \alpha_{11}$

$A_{22} := A_{22} - a_{21} a_{12}^T$

$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \leftarrow \left(\begin{array}{c|cc} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right)$

endwhile

when applied to an $m \times n$ matrix.

- ☒ $mn^2 - \frac{1}{3}n^3$
- ☐ $\frac{2}{3}n^3$
- ☐ $\frac{1}{2}mn^2 - \frac{1}{6}n^3$



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Homework 5.2.2.3

1/1 point (graded)
Implement the algorithm

$$A = \text{LU-right-looking}(A)$$
$$A \rightarrow \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right)$$
$$A_{TL} \text{ is } 0 \times 0$$
$$\text{while } n(A_{TL}) < n(A)$$
$$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c|cc} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right)$$
$$\underline{a_{21} := a_{21} / \alpha_{11}}$$
$$\underline{A_{22} := A_{22} - a_{21} a_{12}^T}$$
$$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \leftarrow \left(\begin{array}{c|cc} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right)$$
$$\text{endwhile}$$

as

```
function [ A_out ] = LU_right_looking( A )
```

by completing the code in [Assignments/Week05/matlab/LU_right_looking.m](#). Input is an m \times n matrix A. Output is the matrix A that has been overwritten by the LU factorization. You may want to use [Assignments/Week05/matlab/test_LU_right_looking.m](#) to check your implementation.

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